

LAB NOC 04

Cisco Catalyst 2960 — STP Loop Detection & Prevention

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware / Hardware	Cisco Catalyst 2960 (WS-C2960-24TT-L)	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Ticket ID	NOC-LAB-04	NOC-LAB-04
Categoría / Category	STP — Detección y prevención de loops L2	STP — Layer 2 loop detection and prevention
Nivel / Level	CCNA — Switching / Spanning Tree Protocol	CCNA — Switching / Spanning Tree Protocol
Duración / Duration	Aprox. 45 minutos	Aprox. 45 minutes
Entorno / Environment	Hardware real (equipo usado)	Real hardware (used equipment)
Relevancia NOC	Diagnóstico de loops L2, análisis de syslog	L2 loop diagnosis, syslog analysis

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Crear un loop de Capa 2 en hardware real (Cisco Catalyst 2960), verificar que STP detecta y bloquea automáticamente el puerto redundante, documentar el incidente en formato NOC, remover el loop y confirmar la reconvergencia de STP mediante análisis de syslog.	Create a Layer 2 loop on real hardware (Cisco Catalyst 2960), verify that STP automatically detects and blocks the redundant port, document the incident in NOC format, remove the loop, and confirm STP reconvergence through syslog analysis.

INCIDENT TICKET — NOC-LAB-04

CAMPO / FIELD	DETALLE / DETAIL
Ticket ID	NOC-LAB-04
Priority / Prioridad	Medium
Service / Servicio	LAN Switching — Layer 2
Device / Equipo	Cisco Catalyst 2960 — Access Layer Switch
Symptoms / Sintomas	Inestabilidad de red, pérdida de paquetes intermitente, tiempos de respuesta lentos / Network instability, intermittent packet loss, slow response times
Initial Hypothesis	Posible loop de Capa 2 — STP podría estar bloqueando un puerto activamente / Possible Layer 2 loop — STP may be actively blocking a

Topology	port Fa0/6 conectado directamente a Fa0/16 en el mismo switch — path redundante / Fa0/6 directly connected to Fa0/16 on the same switch — redundant path
Root Cause	Loop de Capa 2 creado por cable directo Fa0/6 ↔ Fa0/16 / Layer 2 loop created by direct cable Fa0/6 ↔ Fa0/16
STP Action	STP coloco Fa0/16 en estado Blocking para prevenir broadcast storm / STP placed Fa0/16 in Blocking state to prevent broadcast storm
Resolution	Desconexion del cable de Fa0/16 — remocion del path redundante / Disconnection of Fa0/16 cable — removal of redundant path
Status	CERRADO — red estable, STP reconvergado / CLOSED — network stable, STP reconverged

PROCEDIMIENTO — 5 ETAPAS / PROCEDURE — 5 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Baseline: verificar estado inicial del switch	STAGE 1 — Baseline: verify initial switch state
2	show spanning-tree → identificar Root Bridge, roles de puertos y estados actuales	show spanning-tree → identify Root Bridge, port roles and current states
3	show etherchannel summary → verificar estado de bundles preexistentes	show etherchannel summary → verify status of pre-existing bundles
4	show vlan brief → confirmar VLANs activas y asignaciones	show vlan brief → confirm active VLANs and assignments
5	ETAPA 2 — Creacion del loop: conectar cable Fa0/6 ↔ Fa0/16 en el mismo switch	STAGE 2 — Loop creation: connect cable Fa0/6 ↔ Fa0/16 on the same switch
6	Observar sintomas: inestabilidad de red, perdida de paquetes, lentitud / Observe symptoms: network instability, packet loss, slowness	
7	ETAPA 3 — Diagnostico: ejecutar show spanning-tree y confirmar hallazgos	STAGE 3 — Diagnosis: run show spanning-tree and confirm findings
8	Confirmar: Fa0/6 en estado Forwarding (Designated) — Fa0/16 en estado Blocking (Non-Designated)	Confirm: Fa0/6 in Forwarding state (Designated) — Fa0/16 in Blocking state (Non-Designated)
9	Confirmar: STP protejo la red — NO hubo broadcast storm gracias al protocolo	Confirm: STP protected the network — NO broadcast storm occurred thanks to the protocol
10	Ejecutar: show interfaces status, show mac address-table, show logging (metodologia NOC completa)	Run: show interfaces status, show mac address-table, show logging (complete NOC methodology)
11	ETAPA 4 — Resolucion: desconectar cable de Fa0/16 — remover path redundante	STAGE 4 — Resolution: disconnect cable from Fa0/16 — remove redundant path
12	ETAPA 5 — Verificacion: show logging → confirmar LINEPROTO-5-UPDOWN en syslog	STAGE 5 — Verification: show logging → confirm LINEPROTO-5-UPDOWN in syslog

13	Verificar STP reconvergencia — Fa0/6 continua en Forwarding, red estable	Verify STP reconvergence — Fa0/6 continues in Forwarding, network stable
14	Cerrar ticket NOC-LAB-04 — incidente documentado y resuelto	Close ticket NOC-LAB-04 — incident documented and resolved

INCIDENT TICKET

LAB NOC 04 — STP Loop Detection

Ticket ID: NOC-LAB-04

Priority: Medium

Service: LAN Switching

Device: Cisco Catalyst 2960

Location: Access Layer Switch

Problem Description

Multiple users reported intermittent connectivity issues on the local network.

Symptoms observed:

- Network instability
- Occasional packet loss
- Slow response times

Possible Layer 2 issue suspected.

Initial Hypothesis

A potential **Layer 2 loop** might exist in the switch network topology.

Spanning Tree Protocol (STP) may be actively preventing the loop by blocking one of the switch interfaces.

Initial Baseline Commands

The NOC engineer begins troubleshooting with:

```
show vlan brief
show spanning-tree
show interfaces status
```

Expected Network Topology (Baseline)

Switch 2960

Fa0/6 ----- cable ----- Fa0/16

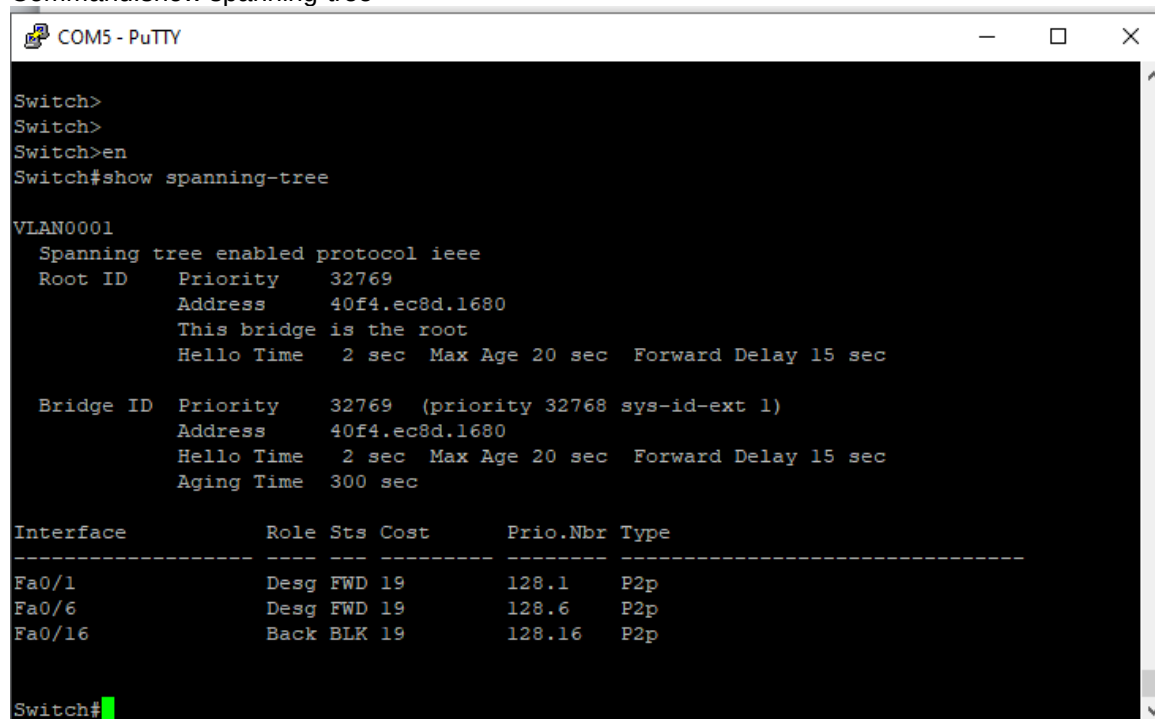
This connection may create **redundant Layer 2 paths**, triggering STP protection mechanisms.

Investigation Goal

Determine:

- Which interface STP places in **blocking state**
- Whether a **loop condition exists**
- If the network is protected by STP

Command: show spanning-tree



```
COM5 - PuTTY
Switch>
Switch>
Switch>en
Switch#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     40f4.ec8d.1680
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     40f4.ec8d.1680
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Desg FWD 19        128.1   P2p
Fa0/6                    Desg FWD 19        128.6   P2p
Fa0/16                   Back BLK 19        128.16  P2p

Switch#
```

Findings

STP detected a potential Layer 2 loop between interfaces Fa0/6 and Fa0/16.

To prevent a switching loop, STP placed interface **Fa0/16 in Blocking state**, allowing traffic to flow only through Fa0/6.

Network remained stable thanks to STP protection.

Cisco 2960

Fa0/6 ----- cable ----- Fa0/16

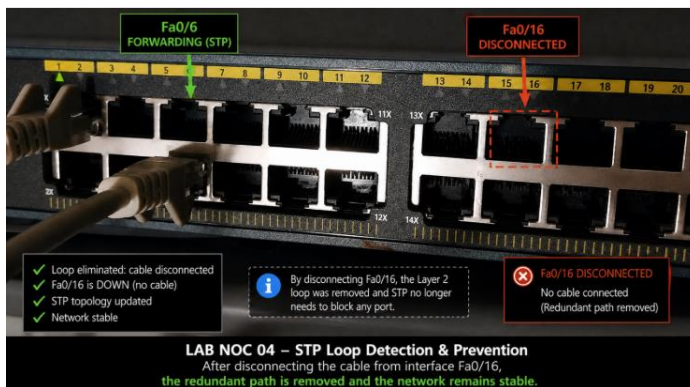
Ese cable crea **un loop dentro del mismo switch**.



```
COM5 - PuTTY
Switch#show interfaces status

Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   1          a-full  a-100  10/100BaseTX
Fa0/2     Fa0/2     notconnect  1          auto    auto   10/100BaseTX
Fa0/3     Fa0/3     notconnect  1          auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1          auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  1          auto    auto   10/100BaseTX
Fa0/6     Fa0/6     connected   1          a-full  a-100  10/100BaseTX
Fa0/7     Fa0/7     notconnect  1          auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  1          auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  1          auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  1          auto    auto   10/100BaseTX
Fa0/11    Fa0/11    notconnect  1          auto    auto   10/100BaseTX
Fa0/12    Fa0/12    notconnect  1          auto    auto   10/100BaseTX
Fa0/13    Fa0/13    notconnect  1          auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  1          auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  1          auto    auto   10/100BaseTX
Fa0/16    Fa0/16    connected   1          a-full  a-100  10/100BaseTX
Fa0/17    Fa0/17    notconnect  1          auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  1          auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  1          auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  1          auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1          auto    auto   10/100BaseTX
--More--
```

Desconecte del switch el cable de la interfaz fastEthernet 0/16. Al desconectar ya no hay camino redundante, STP no necesita bloquear.



```
Switch#
Switch#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     40f4.ec8d.1680
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     40f4.ec8d.1680
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Desg FWD 19        128.1    P2p

Switch#
```

Incident analysis

A Layer 2 loop was created between interfaces **Fa0/6** and **Fa0/16**.

Spanning Tree Protocol automatically placed **Fa0/16 in blocking state** to prevent a broadcast storm.

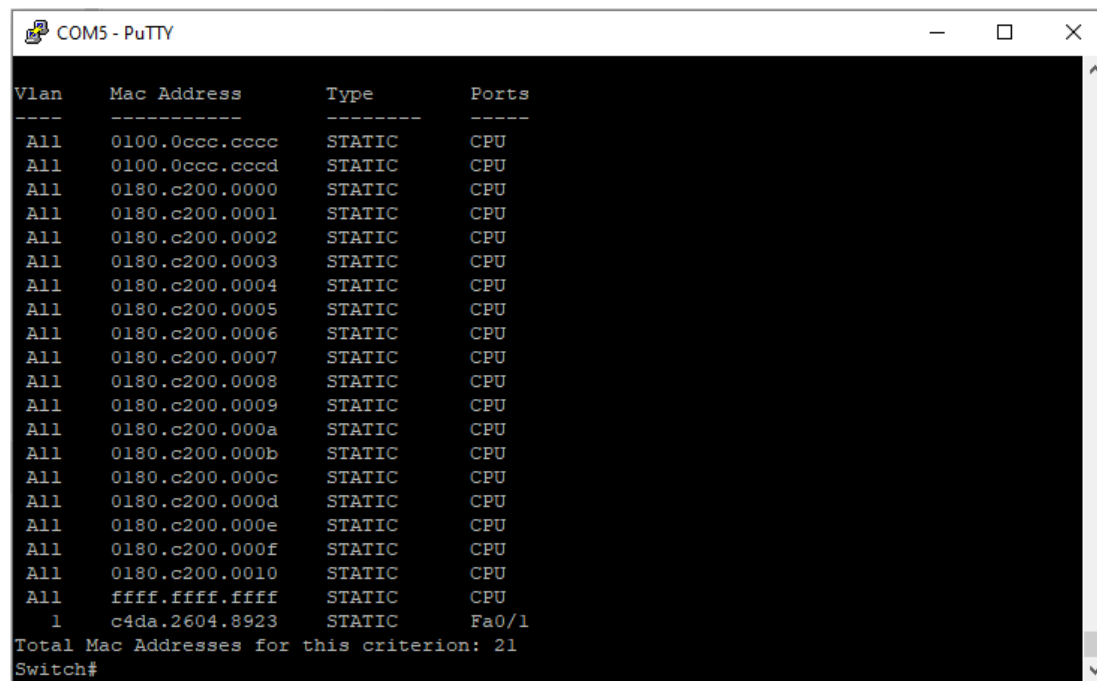
After removing the redundant link, STP reconverged and the remaining active port continued forwarding traffic.

Network operation returned to normal.

Resultado del LAB 04

- ✓ loop detectado
- ✓ STP bloqueó puerto
- ✓ se removió el loop
- ✓ STP reconvergió

Incidente resuelto



```
COM5 - PuTTY

Vlan      Mac Address      Type      Ports
-----
All       0100.0ccc.cccc   STATIC    CPU
All       0100.0ccc.cccd   STATIC    CPU
All       0180.c200.0000   STATIC    CPU
All       0180.c200.0001   STATIC    CPU
All       0180.c200.0002   STATIC    CPU
All       0180.c200.0003   STATIC    CPU
All       0180.c200.0004   STATIC    CPU
All       0180.c200.0005   STATIC    CPU
All       0180.c200.0006   STATIC    CPU
All       0180.c200.0007   STATIC    CPU
All       0180.c200.0008   STATIC    CPU
All       0180.c200.0009   STATIC    CPU
All       0180.c200.000a   STATIC    CPU
All       0180.c200.000b   STATIC    CPU
All       0180.c200.000c   STATIC    CPU
All       0180.c200.000d   STATIC    CPU
All       0180.c200.000e   STATIC    CPU
All       0180.c200.000f   STATIC    CPU
All       0180.c200.0010   STATIC    CPU
All       ffff.ffff.ffff   STATIC    CPU
1         c4da.2604.8923   STATIC    Fa0/1
Total Mac Addresses for this criterion: 21
Switch#
```

Command: show logging

```
*Mar 1 03:05:56.504: %LINK-3-UPDOWN: Interface Port-channel1, changed state to down
*Mar 1 03:06:17.450: %SYS-5-CONFIG_I: Configured from console by console
*Mar 1 03:47:43.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, c
hanged state to down
*Mar 1 03:47:43.121: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, ch
anged state to down
*Mar 1 03:47:44.085: %LINK-3-UPDOWN: Interface FastEthernet0/16, changed state to down
*Mar 1 03:47:44.127: %LINK-3-UPDOWN: Interface FastEthernet0/6, changed state to down
```

Interface FastEthernet0/16 changed state to down

Interface FastEthernet0/6 changed state to down

Lo que hiciste en el LAB 04

El laboratorio era **STP Loop Detection and Prevention**.

Pasos que realizaste:

1 Creaste un loop

Conectaste:

Fa0/6 ----- Fa0/16

Esto genera un **loop de capa 2**.

2 Verificaste STP

Comando usado:

show spanning-tree

Resultado:

- un puerto **Forwarding**
- otro **Blocking**

STP evitó el broadcast storm.

3 Simulaste troubleshooting

Ejecutaste:

```
show interfaces status
show mac address-table
show logging
```

4 Quitaste el cable

Desconectaste: Fa0/16

5 Analizaste los logs

Comando:

```
show logging
```

Los logs mostraron:

```
Interface FastEthernet0/16 changed state to down
Interface FastEthernet0/6 changed state to down
```

Eso permitió **reconstruir el incidente**.

Resultado del laboratorio

- ✓ Loop detectado
- ✓ STP bloqueó un puerto
- ✓ Se removió el cable
- ✓ STP reconvergió
- ✓ Incidente documentado

Eso **cierra el ticket del laboratorio**.

Incident Resolution

A Layer 2 loop was created between interfaces **Fa0/6** and **Fa0/16**.

Spanning Tree Protocol detected the loop and placed one interface in **blocking state**, preventing a broadcast storm.

After disconnecting interface **Fa0/16**, the redundant path was removed and the network returned to normal operation.

Logs confirmed the sequence of link state changes.

ESTADO STP OBSERVADO / OBSERVED STP STATE

PUERTO / PORT	ESTADO STP / STP STATE	ROL / ROLE	SIGNIFICADO / MEANING
Fa0/6	Forwarding	Designated	Trafico fluye normalmente / Traffic flows normally
Fa0/16	Blocking	Non-Designated	STP bloqueo el puerto para evitar loop / STP blocked port to prevent loop

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
<code>show spanning-tree</code>	Muestra el estado STP: Root Bridge, roles y estados de puertos	Shows STP state: Root Bridge, port roles and states
<code>show interfaces status</code>	Muestra estado fisico de cada puerto (connected/notconnect/err-disabled)	Shows physical state of each port (connected/notconnect/err-disabled)
<code>show vlan brief</code>	Verifica VLANs activas y asignacion de puertos	Verifies active VLANs and port assignments
<code>show mac address-table</code>	Tabla MAC dinamica — util para rastrear origen del trafico	Dynamic MAC table — useful for tracking traffic source
<code>show logging</code>	Muestra syslog del switch — clave para reconstruir la secuencia del incidente	Shows switch syslog — key for reconstructing incident timeline
<code>show etherchannel summary</code>	Verifica estado de bundles EtherChannel (contexto de baseline)	Verifies EtherChannel bundle status (baseline context)

EVIDENCIA SYSLOG / SYSLOG EVIDENCE

#	MENSAJE /	LOG OUTPUT	INTERPRETACION /
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	MESSAGE		INTERPRETATION
1	LINEPROTO-5-UPDOWN	Interface FastEthernet0/16 changed state to down	Cable desconectado de Fa0/16 — path redundante removido / Fa0/16 cable disconnected — redundant path removed
2	LINEPROTO-5-UPDOWN	Interface FastEthernet0/6 changed state to down	Fa0/6 tambien bajo (sin destino — loop eliminado) / Fa0/6 also went down (no destination — loop eliminated)
3	STP converge	STP reconverged — Fa0/6 returns to Forwarding	Red estabilizada — unico path activo / Network stabilized — single active path

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
Loop L2 entre Fa0/6 y Fa0/16 causo inestabilidad de red (síntomas reportados por usuarios). / L2 loop between Fa0/6 and Fa0/16 caused network instability (symptoms reported by users). Durante el diagnostico se verifico que STP estaba activo y funcionando correctamente — es facil asumir que STP fallo cuando en realidad lo estaba manejando. / During diagnosis it was verified that STP was active and working correctly — it's easy to assume STP failed when it was actually handling it. Los logs de syslog mostraron la secuencia exacta: Fa0/16 down → Fa0/6 down → reconvergencia. Permiten reconstruir el timeline del incidente. / Syslog showed the exact sequence: Fa0/16 down → Fa0/6 down → reconvergence. Allows reconstructing the incident timeline.	STP detecto el loop automaticamente y coloco Fa0/16 en Blocking. Red no colapso. No requirio intervencion manual inmediata. / STP automatically detected the loop and placed Fa0/16 in Blocking. Network did not collapse. No immediate manual intervention required. Regla NOC: antes de intervenir, leer el estado de STP completo. Si un puerto esta en Blocking es porque STP hizo su trabajo — no se reactiva sin investigar la causa del loop. / NOC rule: before intervening, read the full STP state. If a port is Blocking, STP did its job — do not reactivate without investigating the loop cause. show logging es evidencia forense del incidente. En NOC siempre se captura antes de cerrar el ticket — es la prueba de que el problema ocurrio y fue resuelto. / show logging is forensic evidence. In NOC it is always captured before closing the ticket — it proves the problem occurred and was resolved.

RESULTADO / RESULT

RESULTADO (ES) Loop L2 detectado. STP bloqueo Fa0/16 automaticamente. Cable removido. STP reconvergió. Syslog analizado y documentado como evidencia forense. Ticket NOC-LAB-04 cerrado.	RESULT (EN) L2 loop detected. STP automatically blocked Fa0/16. Cable removed. STP reconverged. Syslog analyzed and documented as forensic evidence. Ticket NOC-LAB-04 closed.
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KEY TAKEAWAY

"STP no es un problema — es la solución. Cuando un puerto esta en Blocking, el switch esta

protegiendo la red de un broadcast storm. En NOC, antes de 'arreglar' un puerto bloqueado hay que entender POR QUE esta bloqueado. El syslog es la evidencia forense que permite reconstruir el timeline del incidente y cerrar el ticket con pruebas concretas."

"STP is not the problem — it is the solution. When a port is in Blocking state, the switch is protecting the network from a broadcast storm. In NOC, before 'fixing' a blocked port you must understand WHY it is blocked. The syslog is the forensic evidence that allows reconstructing the incident timeline and closing the ticket with concrete proof."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining environment (e.g., Newmont Cerro Negro), L2 loops are one of the most dangerous network events — a broadcast storm can saturate the entire LAN segment in seconds, taking down SCADA communications, IP cameras, and safety systems simultaneously. A NOC engineer must: (1) recognize the symptoms (high CPU on switch, broadcast traffic spikes), (2) verify STP is active and blocking the correct port, (3) identify the physical cause of the redundant path (miscabling, unauthorized device), (4) remove the loop source, and (5) document the full incident timeline using show logging before closing the ticket. Never re-enable a blocked port without understanding why STP blocked it.